



remember from AE 319, the supersonic jet can have a pressure different from ambient ($P_e \neq P_a$). we can show $\tau = \textcircled{1} + \textcircled{2}$

$$\tau = \underbrace{\textcircled{1}}_{\text{thrust force}} + \underbrace{\textcircled{2}}_{\text{pressure difference between exit jet \& ambient}}$$

$\tau \begin{cases} \textcircled{1} \rightarrow m u_e : \text{Momentum of exit jet} \\ \textcircled{2} \rightarrow \text{pressure difference between exit jet \& ambient} \end{cases}$

usually $\textcircled{1} \gg \textcircled{2}$

if the exit flow is subsonic, $P_e = P_a \Rightarrow \textcircled{2} = 0$

" " " " " perfectly expanded $\Rightarrow \textcircled{2} = 0$

" " " " " underexpanded $\Rightarrow P_e > P_a \Rightarrow \textcircled{2} > 0$

" " " " " over $\Rightarrow P_e < P_a \Rightarrow \textcircled{2} < 0$

$\tau = m u_e + (P_e - P_a) A_e$, we can write: $\tau = m u_{eq}$

$$u_{eq} = u_e + \frac{(P_e - P_a) A_e}{\dot{m}} = u_e + \frac{(P_e - P_a) A_e}{\rho_e u_e A_e}$$

(P-1)

$$u_{eq} = u_e + \frac{P_e - P_a}{\delta_e u_e} = u_e + \frac{P_e}{\delta_e u_e} - \frac{P_a}{\delta_e u_e} = u_e + \frac{\cancel{\delta_e} R T_e}{\cancel{\delta_e} u_e} - \frac{P_a}{\delta_e u_e}$$

$$\Rightarrow u_{eq} = u_e + \frac{R T_e}{u_e} - \frac{P_a}{\delta_e u_e} = u_e + \frac{R T_e}{u_e} \left(1 - \frac{P_a}{\cancel{R T_e \delta_e} P_e}\right)$$

$$\Rightarrow u_{eq} = u_e + \frac{R T_e}{u_e} \left(1 - \frac{P_a}{P_e}\right)$$